

Modelling and Simulation of Fast Autonomous Exploration and Mapping System for GPS Denied Unknown Environment Using ROS

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Abstract— This paper describes a multi-robot coordinated heterogeneous vehicle which employs cooperative unmanned ground vehicles (UGVs) and unmanned aerial vehicles (UAVs) to explore and map a GPS-denied unknown terrain. Exploration processes like cave exploration, tunnel inspections required an accurate and quick mapping of the environment in emergencies for rescue or maintenance operations. This multi-robot system's main objective is to use the Dijkstra algorithm to produce a 2D and 3D map of an unexplored region. Each robot features a border identification system that uses image processing to derive border locations. The autonomous exploration problem is a partially observable Markov decision process (POMDP) to the local planner exploration component. The robot is directed to the uncharted region in the map after recovering boundary points in the region separating known from unknown space. The global planner graph, introducing flexible values, is constructed during the discovery phase to examine the robot's border positions at the end. The two methodologies are combined with speeding up the search for border sites. Closed-loop control ensures that the robot is always pushed to its greatest advantage following achieving the desired advantages. Finally, the 2D and 3D maps obtained from each robot are combined using conventional map merging techniques to form a map.

Keywords— Autonomous Exploration, Path planning, Dijkstra algorithm, Collaborative system, heterogeneous vehicular system.

I. INTRODUCTION

Autonomous robots such as unmanned ground vehicles (UGVs) and unmanned aerial vehicles (UAVs) are becoming a new adoption in various fields of real-world applications. The primary reason behind this faster implementation of such autonomous robots in applications is their ability to move around an unknown complex environment. It can be able to execute the desired task without any human hindrance. Many existing robots need a well-developed environment model to move around and accomplish the desired task. Without these predefined navigation models, the robots are equipped with a GPS-based external environment map. In this work, we focus on developing an autonomous exploration of a coordinated robot system in an entirely GPS-denied, wholly unknown environment and construct the 2-dimensional (2D) occupancy grid map and 3-dimensional (3D) point cloud data to map the unknown workspace effectively.

Some existing exploration robots require a piece of partial information about the environment to plan their path [1]-[4]. But those frameworks fail in a completely unknown environment. To avoid this, existing systems use an active SLAM strategy that does not require any information about

the environment to plan the robot path [5]-[7]. However, the framework is limited to specific applications as this technique drives the robots to acquire the possible knowledge of the entire environment, which the existing system does not require. Therefore, a combination of suitable exploration and various applications uses multiple robots to carry out the desired task cooperatively [8]-[12].

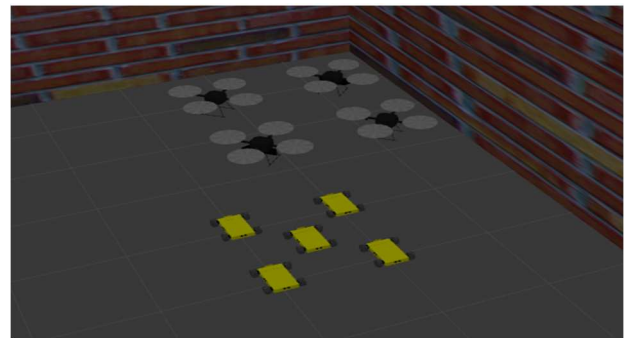


Fig.1. Multi-Robot system in ROS - Gazebo framework

Implementing collaborative tasks enables the multi-robot system to share the information between the robots in the system, thus decomposing the entire environment exploration and mapping efforts between the individual robots. The distribution of the effort between the robots in the system increases the overall speed of operation by increasing the number of robots in the system and flexibility, enabling the system to reach more complex parts of the environment and map it. Generally, UAVs have lesser endurance in their flight time, which affects the entire system's operational time. The distributed load can reduce the UAVs' load, increasing the system operation time. Applications like search and rescue operations, automated inventory management, and tunnel surface inspection activities where the human presence cannot require immediate action under emergency conditions. Multi-robot systems enhance the desired objective of exploring and mapping the unknown environment more quickly than the single robot's work time.

This paper proposes an autonomous exploration and mapping system using 5 UGVs and 4 UAVs in heterogeneous teams. A heterogeneous vehicular system is approached to overcome constraints like ability to map a vast volume of environment. The proposed system follows the leader-follower technique to plan the navigation of the entire system using frontier points. The leader UGV is placed at the centre of the square, wherein the four followers UGVs are placed at

the corners of the square with a predefined distance between them. Once the frontier points are created, the leader UGV plans the exploration path avoiding the obstacles using the Dijkstra algorithm. The follower UGVs with the formation control follow the leader robot and create the environment's 2-dimensional occupancy grid map. Each follower UGV in the system is followed by a UAV at a specific height. The follower UGVs are equipped with a LIDAR sensor for creating the 2D map. Each UAV is equipped with a depth sensor camera that creates the point cloud data of the environment and enables real-time image data of the environment for real-time or post-processing purposes. For the maximum field of coverage, the depth sensor camera is at an angle with respect to the system's frame. The 2D map from each UGV is merged simultaneously during the mapping process, whereas the point cloud data acquired from each UAV is processed externally and merged using external software. Using the proposed multi-robot collaborative perception strategy, we demonstrate the autonomous exploration and mapping of a GPS – denied unknown environment without any pre-planned trajectories.

Nexus omnidirectional robots for unmanned ground vehicles and Hector quadrotors for unmanned aerial vehicles are used to test the proposed multi-robot system in the Robot Operating System (ROS) – Gazebo environment. ROS is an open-source framework that helps the robotics community to build and test the algorithms developed using virtual robotic tools. In ROS, a single task is broken down into many fine-grained scale tasks, and each of them is carried out in a node. ROS nodes combine and communicate between them in a graph using ROS topics. ROS nodes can be monitored using the Node tools available in the ROS framework.

II. SYSTEM DYNAMIC MODEL AND CONTROL ALGORITHM

A. Omnidirectional mobile robot dynamics

When compared to other wheeled robots, omnidirectional wheeled robots have unique dynamics. The wheels on these robots are fixed standard wheels with passive rollers around the perimeter. The angle between the passive roller axis rotation and the plane of the wheel on this Mecanum wheel is 45 degrees. The geometric centre of the mobile robot is G, and ${}^I r_G$ is the position vector concerning the inertial frame $\{I\}$. In the robot body frame $\{B\}$, G' represents the centre of mass of the mobile platform. The relative position vector of the moving platform ${}^B r_{G'/G} = [-d_1 \ -d_2]^T$ in body frame $\{B\}$. The velocity ${}^B V_G$ of point G concerning robot body frame $\{R\}$ is expressed as

$${}^B V_G = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} \dot{x}_1 \\ \dot{y}_1 \end{bmatrix} \quad (1)$$

where $\dot{x}_1, \dot{y}_1$ are velocity components along X and Y axes correspondingly and θ is the angle at which the robot platform is orientated with reference to the inertial frame. The velocity of the centre of mass of the platform with reference to the body frame is given as

$${}^B V_{G'} = {}^B V_G + (\dot{\theta} k_B \times {}^B r_{G'/G})$$

$$= (\dot{x}_1 \cos(\theta) + \dot{y}_1 \sin(\theta) + \dot{\theta} d_2) i_B + (-\dot{x}_1 \sin(\theta) + \dot{y}_1 \cos(\theta) + \dot{\theta} d_2) j_B \quad (2)$$

The equation of motion of each omnidirectional Mecanum wheeled robot is expressed in matrix form as [13]

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + B^T S f = \frac{1}{r} B^T \tau \quad (3)$$

M denotes the Generalized Mass matrix and is obtained by the following matrix $M_{3 \times 3}$

$$\begin{aligned} M_{11} &= M_{22} = m_b + 4 \left(m_w + \frac{I}{r^2} \right); \\ M_{13} &= M_{31} = m_b (d_1 \sin \theta + d_2 \cos \theta); \\ M_{23} &= M_{32} = m_b (-d_1 \cos \theta + d_2 \sin \theta); \\ M_{12} &= M_{21} = 0; \\ M_{33} &= m_b (d_1^2 + d_2^2) + I_b + 8(m_w + \frac{I}{r^2}) l^2 \sin^2(\frac{\pi}{4} - \alpha); \end{aligned} \quad (4)$$

C is obtained by the following matrix $C_{3 \times 3}$

$$\begin{aligned} C_{11} &= C_{12} = C_{21} = C_{22} = C_{31} = C_{32} = C_{33} = 0; \\ C_{13} &= m_b \dot{\theta} (d_1 \cos \theta - d_2 \sin \theta); \\ C_{23} &= m_b \dot{\theta} (d_1 \sin \theta + d_2 \cos \theta); \end{aligned} \quad (5)$$

B denotes the Geometric Jacobian Matrix and is obtained by the following matrix $B_{4 \times 3}$

$$\begin{aligned} B_{11} &= -(\cos \theta - \sin \theta); \quad B_{12} = -(\cos \theta + \sin \theta); \\ B_{21} &= -(\cos \theta + \sin \theta); \quad B_{22} = -(\sin \theta - \cos \theta); \\ B_{31} &= (\cos \theta - \sin \theta); \quad B_{32} = (\sin \theta + \cos \theta); \\ B_{41} &= (\cos \theta + \sin \theta); \quad B_{42} = (\sin \theta - \cos \theta); \\ B_{14} &= B_{24} = B_{34} = B_{44} = -\sqrt{2} l \sin(\frac{\pi}{4} - \alpha); \end{aligned} \quad (6)$$

S matrix is obtained by the following matrix $S_{4 \times 4}$

$$\begin{aligned} S &= \text{diag}(\text{sgn}(\dot{\phi}_1) \text{sgn}(\dot{\phi}_2) \text{sgn}(\dot{\phi}_3) \text{sgn}(\dot{\phi}_4)) \\ \begin{bmatrix} \dot{\phi}_1 \\ \dot{\phi}_2 \\ \dot{\phi}_3 \\ \dot{\phi}_4 \end{bmatrix} &= -\left(\frac{\sqrt{2}}{r}\right) \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 & l \sin(\frac{\pi}{4} - \alpha) \\ \sqrt{2}/2 & -\sqrt{2}/2 & l \sin(\frac{\pi}{4} - \alpha) \\ -\sqrt{2}/2 & -\sqrt{2}/2 & l \sin(\frac{\pi}{4} - \alpha) \\ -\sqrt{2}/2 & \sqrt{2}/2 & l \sin(\frac{\pi}{4} - \alpha) \end{bmatrix} \\ &\cdot \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{x}_1 \\ \dot{y}_1 \\ \dot{\theta} \end{bmatrix} \end{aligned} \quad (7)$$

τ denotes the angular velocity of each wheel and is obtained by the following matrix $\tau_{4 \times 1}$

$$\tau = [\tau_1 \ \tau_2 \ \tau_3 \ \tau_4]^T \quad (8)$$

f denotes the frictional force on each wheel of the robot and is obtained by the following matrix $f_{4 \times 1}$

$$f = [f_1 \ f_2 \ f_3 \ f_4]^T \quad (9)$$

B. Quadrotor dynamics

The four motors operate the quadrotors, and the thrust produced by each motor determines the system's movement. The input control of the quadrotor is $[x, y, z, \dot{x}, \dot{y}, \dot{z}, \phi, \theta, \varphi, \dot{\phi}, \dot{\theta}, \dot{\varphi}]$. The quadrotor nonlinear dynamics are expressed in the state space equation [14-17].

$$\begin{aligned} \ddot{x} &= \frac{1}{m}(\cos\phi\sin\theta\cos\varphi + \sin\phi\sin\varphi)u_4; \\ \ddot{y} &= \frac{1}{m}(\cos\phi\sin\theta\sin\varphi - \sin\phi\cos\varphi)u_4; \\ \ddot{z} &= -g + \frac{1}{m}(\cos\phi\cos\theta)u_4; \\ \ddot{\phi} &= \frac{I_y - I_z}{I_x} \dot{\theta}\dot{\varphi} + \frac{I_r}{I_x} \dot{\theta}\omega_T + \frac{l_q}{I_x} u_1; \\ \ddot{\theta} &= \frac{I_x - I_z}{I_y} \dot{\phi}\dot{\varphi} + \frac{I_r}{I_y} \dot{\phi}\omega_T + \frac{l_q}{I_y} u_2; \\ u_1 &= \omega_4^2 - \omega_2^2; \\ u_2 &= \omega_3^2 - \omega_1^2; \\ u_3 &= \omega_2^2 + \omega_4^2 - \omega_1^2 - \omega_3^2; \\ u_4 &= \omega_1^2 + \omega_2^2 + \omega_3^2 + \omega_4^2; \\ \omega_T &= \omega_2 + \omega_4 - \omega_1 - \omega_3; \end{aligned} \quad (10)$$

where ω_i is the angular velocity of the quadrotor's i^{th} rotor motor, l_q is the length of the quadrotor's wing, m is the quadrotor's mass, g is the gravitational force, and I_x, I_y, I_z are the moment of inertia around X, Y, Z axes, respectively.

C. Controller Design

A Proportional Integral Derivative (PID) controller is used to operate the suggested system, which is manually adjusted to achieve the greatest potential process efficiency. The ROS node is used to distribute the controller algorithm to the system. Table 1 shows the PID values that were utilised in the system.

Table I: PID values for the proposed system

	Ground Robot	Quadcopter
P	0.88	1.5
I	0.4	0.5
D	0	0

III. SYSTEM WORKFLOW AND ALGORITHM

In an unknown environment exploration system, the system should begin by autonomously exploring and mapping the environment to determine how to begin its operation. In a SLAM approach, autonomous exploration and map creation are usually two separate processes. During the movement, the Frontier-Based method for autonomous exploration algorithm calculates the system's goal position. In contrast, the Gmapping algorithm creates the 2D occupancy map using the information from the LIDAR scanned data. This process continues, and the map is updated regularly when the robot reaches an unknown area. The system workflow is shown in Fig.2.

After obtaining the map, the Dijkstra path planning algorithm uses the generated map as input to execute global path planning. The DWA planner will avoid environmental obstacles by using the laser scan data acquired from the LIDAR sensor. The system will plan a collision-free shortest path to the target point by combining these two path planners.

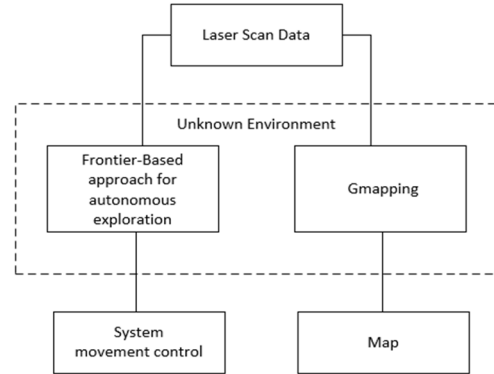


Fig.2. System workflow for unknown map

A. Global Path Planning Algorithm

The grid-based map is used in this work to process the completely unknown environment. The entire environment is divided into identical uniform cells in a grid-based map. Each cell in the environment is assigned a value based on the likelihood of encountering an obstacle. This type of grid-based map isn't suitable for large environments. To overcome this, a suitable algorithm for global path planning should be chosen, which should be more efficient in terms of timeliness and return the shortest path.

The Dijkstra algorithm is a popular single-source path planning method that finds the shortest path between the start and goal nodes. Using the weights of the edges, the Dijkstra algorithm produces the shortest path between the source and target nodes. Breadth-first Search (BFS) can compute the cost of any given node to the target node. BFS involves selecting the nearest node to the target node instead of the closest node to the source node. Dijkstra algorithm centres the source node and extends outwards until the target node is reached. Therefore, the Dijkstra algorithm is chosen as the system's global path-planning method.

B. Local Path Planning Algorithm

Obstacle avoidance is major consideration which the multi-robot system should consider. The Dynamic Window Approach (DWA) obtains the global plan and cost map and then produces suitable velocity instructions for the system to navigate around environmental obstacles. DWA creates a velocity vector space that includes the system's translational and rotational velocities. A set of speeds is created and simulated in that particular instant based on the input. The system drivers pick and process the optimal velocity after the results are assessed from numerous sets.

C. Autonomous Exploration

An autonomous exploration system's major task is navigating the system in a completely unknown environment. The system should navigate from an unoccupied region into an unknown area rather than navigating again in the explored area. The edge grid is the empty grid that exists next to any unknown grid. The system should navigate to the edge grid,

expanding the exploration range and building a map of the unknown region.

The tasks in autonomous exploration may be divided into two parts.

1. **Edge Detection:** The edge between the open and unknown regions is determined through edge detection. The region is deemed accessible if the length of the detected edge exceeds the size of the system footprint, else the edge is considered inaccessible.

2. **Target Point Generation and Navigation:** Once an available path has been discovered using edge detection, the system must now build a path from its present state to the target region. At the same time, the system should figure out the shortest route between its present location and the accessible target area. It is a single source with multiple target scenarios. We employ the Dijkstra method to determine the shortest path between the present point and the target region. When the system reaches the target location using the shortest path created by the Dijkstra algorithm, it adds that point to the list of covered edges, which allows the system to go to new unexplored areas rather than mapping the same region again and over.

D. Mapping

Gmapping is used to create 2D occupancy grid maps from LIDAR scan data in this study. The two most prevalent algorithms for SLAM and map generation are Gmapping and Hector slam. Gmapping employs both LIDAR data and odometry values, and it outperforms the Hector SLAM, which solely uses laser scan data. Our suggested system comprises one leader ground robot and four follower ground robots. Each of the four following ground robots has a LIDAR sensor that produces LIDAR scan data and local maps. Local maps from each robot are sent to the ROS nodes using the conventional map merging approach; a centralised map is created from the input. The `multirobot_map_merge` package influenced the map merging package used in the system. Our package carries the map merging procedure online between the follower ground robots.

IV. SIMULATION AND RESULT

This proposed system is developed and simulated in ROS – Gazebo environment. In this article, the NEXUS Omni 4 - Wheeled Mecanum robot is utilised for ground robots, while the Hector Quadrotor is used for aerial robots. All following ground robots include a LIDAR sensor, which is used to create LIDAR scan data, and each aerial robot has a Kinect camera fitted for producing point cloud data.

The leader ground robot is in charge of avoiding obstacles in the environment and planning the system's path. The follower rovers are designed to follow the leader ground robot over a 2-meter distance, forming a square with the leader robot in the middle. A quadrotor with a 2-meter offset above each follower ground robot follows it. The 8-follower robots form a cube configuration with a 2-meters side.

A. Autonomous exploration test

The multi-robot system produces a goal point to advance the system to the undiscovered area based on the frontier points created. The map created by the system is displayed in the figure after a 20-second interval. The white region on the environment represents the explored area of the provided environment. The red arrow in the figure shows the target

position of the system at that time. The target position of the leader robot is provided by the path planner, and the formation control is deployed by an external python script.

B. Indoor Navigation test

The Navigation stack is responsible for creating the frontier points in the unexplored area. The green curve in figure 4 shows the global path, and the small red curve denotes the local path of the system. Figure 4 depicts the system with obstacles on its way. The system successfully replans its path and avoids obstacles while calculating the shortest path to the current destination point. When a local deadlock occurs, the system spins clockwise to re-establish a suitable path for subsequent navigation. This procedure is repeated when the system reaches the target point, and a new set of frontier points is produced to navigate further. The system will not create frontier points if the whole area has been explored, and the search will be terminated.

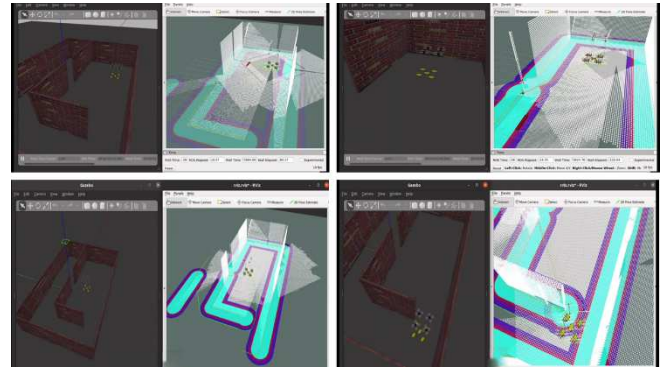


Fig.3. 2D & 3D Mapping of Unknown Environment using the Coordinated UAVs and UGVs with Gazebo and RViz

C. Mapping

The ROS native gmapping package maps the unknown environment in this proposed system. As a ROS node, this module offers laser-based SLAM. The 2D occupancy grid map is created using laser input. Each follower ground robot in the system is equipped with a laser sensor that generates a 2D occupancy map and combines it with the others in real time to map the unknown environment faster. Each quadrotor in the system has a stereo camera mounted at a 45-degree angle to cover the surrounding region effectively. The point cloud data of the environment is produced using the data from the stereo camera. Offline, using an external processor, this point cloud data are processed and combined to create a full 3d map of the unknown area.

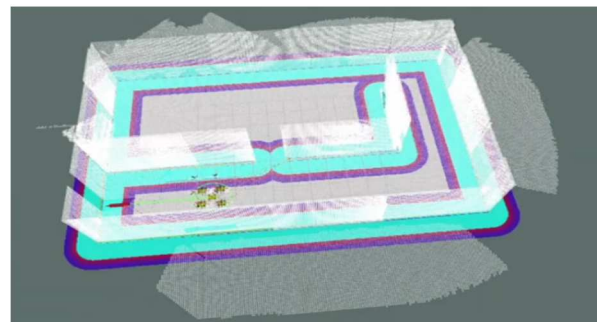


Fig.4. Complete Environment Map of the environment

A single ground robot and a single quadrotor each map the environment separately. The ground robot took 316

seconds to create a 2D occupancy map of the area with an 87 per cent mapping efficiency, while the quadrotor took 248 seconds to create point cloud data with a 74 per cent mapping efficiency. The simulation is completed after 94-95 per cent of the environment has been covered. The proposed system creates a 2D and 3D map of the environment in just 196 seconds, with 89 per cent and 82 per cent efficiency in 2D and 3D mapping, respectively.

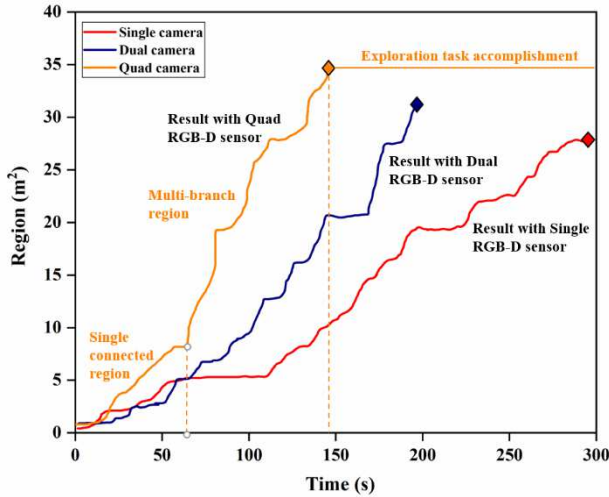


Fig. 5. The comparison of the time-region curves for single, dual, and quad RGB-D sensor configuration in autonomous mapping and exploration

Fig. 5 shows the time region curve of an autonomous mapping and exploration procedure. The resulting graph compares single, dual, and quad-effective RGB-D sensors. While the dual RGB-D sensor system had a 31.3 m exploration job with a period of 196 s, the single RGB-D camera successfully explored the territory of 27.9 m in 297 s. In 148 seconds, the suggested quad RGB-D sensor configuration explored 34.1 m. The view ranges of the single and twin sensor setups were comparable in the single-connected region. As a result, there was little variation in the exploration course. Compared to the single and dual sensor setups, the exploration progress in the single-linked region in the suggested quad-sensor arrangement is significantly higher. The quad camera setup's exploration cost time in the multi-branch areas was noticeably shorter than that of the other two examined systems. Compared to the single and dual camera setups, the quad camera system provided more information about the surroundings in the form of a point cloud. This demonstrates the system's effectiveness.

V. CONCLUSION

This paper offers a new method of integrating ground and aerial robots to tap a completely unknown area faster, including detecting obstacles in the environment, identifying appropriate places for the system to travel without clashing with the obstacles, and route planning and mapping of the environment. It is also observed that the distribution of load among the robots improvise the system and mapping efficiency.

The planner of the envisioned system has a maximum velocity along x and y to be 1.5 m/s and 1.3 m/s, respectively, and a minimum velocity along x and y to be 1.1 and 1.05, respectively. The angular velocity maximum and minimum are set to be 1.82 rad/s and 0.9 rad/s, respectively. The proposed system is mapped using a combination of robot

composition and the envisioned system. The results of that are shown in table II.

Table II: Comparison of Mapping efficiency and Time taken for mapping 90% of the environment

System Composition	Time Taken (sec)	Mapping Efficiency
Single Ground Robot and Single Quadrotor	297	69%
Single Ground Robot and two Quadrotors	196	70.33%
Single Quadrotor and two Ground Robot	121	72%
Proposed System	148	85.14%

The suggested approach is found to substantially decrease the time needed substantially stance traversed by the proposed system to map an environment when comparing the closest frontier exploration algorithm. The route followed by the suggested system throughout the mapping process is also found to be simpler and more efficient, thus demonstrating the efficacy of the proposed exploration method. We separated the load between the eight robots to reduce the mapping time. In the current work, the leader ground robot is responsible for obstacle detection and path planning for the entire system. In the future, each robot will be equipped with a stereo sensor to process the data, find the obstacles, and plan the path accordingly to avoid those obstacles.

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